Tertiary Infotech Academy Pte Ltd

UEN: 201200696W

LEARNER GUIDE

For

Infographics and Data Visualization with PowerPoint

TGS Ref No: TGS-2026064178

Conducted by

Tertiary Infotech Academy Pte Ltd

UEN: 201200696W

Version 4.1

© 2026 Tertiary Infotech Academy Pte Ltd. All rights reserved. · www.tertiarycourses.com.sg

DOCUMENT VERSION CONTROL RECORD

Version Number	Effective Date of Release	Summary of Included Changes	Author
2.0	8 January 2025	Previous approved Learner Guide release.	Tertiary Infotech Academy Pte Ltd
3.0	15 August 2026	Rebuilt from the approved v10 deck with current Microsoft infographic-maker and Copilot guidance, eight progressive activities, detailed PowerPoint procedures and editable evidence packs.	Tertiary Infotech Academy Pte Ltd
4.0	16 August 2026	Added the AI-first data-to-insight-to-infographic workflow, Microsoft 365 Copilot and Gemini Notebook procedures, PowerPoint conversion and comparison, prompt patterns, mock multi-source data, generated practice assets and verification controls. Migrated the package to current course code TGS-2026064178.	Tertiary Infotech Academy Pte Ltd
4.1	16 August 2026	Removed the obsolete learner portal slide at client direction and synchronized the Learner Guide with the revised 122-slide courseware package.	Tertiary Infotech Academy Pte Ltd

© 2026 Tertiary Infotech Academy Pte Ltd. All rights reserved. · www.tertiarycourses.com.sg

TABLE OF CONTENTS

How to Use This Guide **4**
Learning Outcomes **4**
Infographic Foundations **4**
AI-First Data-to-Infographic Workflow **5**
PowerPoint Design System **6**
Microsoft 365 Copilot and Gemini Notebook **7**
Review and Resource Decisions **8**
LU1 — Introduction to Infographics **8**
Activity 1 — Use AI to Clean Data and Extract Insights 8
Activity 2 — Prompt an Evidence-Led Infographic Brief 11
LU2 — Create Infographics in PowerPoint **13**
Activity 3 — Build from Microsoft Infographic Maker 13
Activity 4 — Generate Visual Assets and Refine Them in PowerPoint 15
Activity 5 — Use AI Analysis to Build a Native PowerPoint Data Story 18
Activity 6 — Compare Microsoft 365 Copilot and Gemini Notebook 20
LU3 — Review Infographics **23**
Activity 7 — Peer Coaching and Story Review 23
Activity 8 — Budget and Effectiveness Decision 25
Quick PowerPoint Reference **27**
Assessment **28**
Assessment Flow **28**
Sources and Further Reading **28**
Support **28**

© 2026 Tertiary Infotech Academy Pte Ltd. All rights reserved. · www.tertiarycourses.com.sg

How to Use This Guide

This guide supports Infographics and Data Visualization with PowerPoint (TGS-2026064178), a one-day WSQ course aligned to Infographics and Data Visualisation (TOU-DAT-4004-1.1-1). Use it with the visual course deck and the individual activity folders. The slides explain concepts and show worked examples; this guide carries the detailed procedures. Each activity folder is self-contained with prompts, mock CSV/Excel data, PowerPoint files and, where relevant, generated assets or a Gemini Notebook source pack.

PowerPoint and Copilot interfaces can change. Follow the intent and verify the current control name if your version differs. Use only approved accounts and synthetic or approved data in AI services.

Before class:

- Install or update Microsoft PowerPoint and Excel or confirm Microsoft 365 web access.
- Download the complete activities folder from the LMS or GitHub repository.
- Confirm that every activity folder contains INSTRUCTIONS.md, CHECKLIST.md, PROMPTS.md, CSV, XLSX and PPTX files.
- Keep the original evidence files unchanged; save a named working copy for each activity.

Learning Outcomes

- LO1: Draw insights from infographics to make data-driven decisions.
- LO2: Leverage PowerPoint infographics for articulating data insights to stakeholders.
- LO3: Assess the effectiveness of PowerPoint infographics based on allocated resources in accordance with business goals.

Infographic Foundations

An infographic combines concise text, data and visual structure to help an audience understand important information and relationships. A useful infographic is not merely attractive; it answers a question and supports an appropriate decision or action.

Audience question	Useful infographic form	Typical evidence
What changed over time?	Chronology or timeline	Dates, milestones, trend values
How does it work?	Instructional process, cycle or roadmap	Stages, sequence, inputs and outputs
Which option differs?	Comparison or statistical infographic	Comparable metrics and consistent units
How are parts organised?	Hierarchy, tree or mind map	Levels, categories and relationships

© 2026 Tertiary Infotech Academy Pte Ltd. All rights reserved. · www.tertiarycourses.com.sg

What numbers matter?	Statistical dashboard or big-number story	Definitions, denominators, timeframe and source

Insight extraction

- What it shows — State visible values, differences, trends and exceptions.
- What it may mean — Offer a plausible interpretation without implying unsupported causation.
- Why it matters — Connect the pattern to the audience's business question.
- What to do — Recommend a proportionate next step and name uncertainty or evidence gaps.

AI-First Data-to-Infographic Workflow

The default modern workflow is not to construct a full infographic manually from a blank slide. Use approved AI tools to accelerate source organisation, data cleanup, analysis, insight extraction and first-draft visual generation. Use Excel and the original sources to verify evidence, then use PowerPoint as the editable refinement, presentation and release layer.

Collect — Approved files, URLs, transcripts and datasets; record origin and access date.

Clean — Remove duplicates, standardise units and labels, identify missing values and retain the raw source.

Analyse — Ask AI for patterns, exceptions and comparisons, then reproduce critical calculations in Excel.

Extract — Write show–mean–matter–act insights with citations, caveats and forbidden claims.

Generate — Create infographic or slide-deck options from the verified brief and source pack.

PowerPoint — Download or rebuild in PPTX; restore editability, charts, source labels, brand and accessibility.

Verify — Compare every claim to source, record prompts and human changes, approve and export.

Data and insight verification gate

- Keep the untouched raw source and create a source register before cleanup.
- Require AI to list proposed transformations, missing data, conflicting definitions and assumptions.
- Reproduce decision-critical totals, percentages, differences and trends with Excel formulas.

© 2026 Tertiary Infotech Academy Pte Ltd. All rights reserved. · www.tertiarycourses.com.sg

- Separate observation from interpretation; add caveats, confidence and forbidden claims.
- Treat AI as an assistant, not as the source of truth or final owner.

Copy-ready prompt patterns

Data analyst — Using only the attached approved sources, create a source register, identify duplicates or conflicting definitions, clean the table, show formulas or transformations, and return five candidate insights with citations, caveats and a confidence note. Do not invent missing values.

Infographic — Create a one-page executive infographic for Singapore retail managers. Lead with one decision insight, show three supporting metrics, include a five-stage flow and a source footer. Use a clean professional style, strong hierarchy and no invented data. Leave critical labels editable.

Text to image — Create a clean vector-like sustainability delivery illustration for a PowerPoint infographic: one electric van, one reusable parcel, a simple route and a feedback symbol. Transparent background, consistent blue/teal/violet/amber palette, no text, no logo, no watermark.

Gemini Notebook deck — Create Presenter Slides for business stakeholders. Focus on the verified decision insight, use a concise professional style, include source-grounded comparisons and a final recommendation. Avoid claims not supported by selected sources. Download PowerPoint when available; otherwise download PDF and identify what must be rebuilt as editable native PowerPoint objects.

PowerPoint Design System

Design decision	Use in PowerPoint	Quality check
Canvas	Design > Slide Size > Custom Slide Size	Output orientation and channel are defined
Grid	View > Ruler, Gridlines and Guides	Margins and spacing are consistent
Shapes	Insert > Shapes; merge, align, distribute and group	One visual grammar supports the relationship
Icons and images	Insert > Icons or Stock Images	Relevant, consistent, labelled and permitted
Typography	One heading family, one body family, few sizes	Readable hierarchy; no cramped paragraphs
Colour	Restrained palette with one purposeful accent	Meaning does not rely on colour alone
SmartArt	Process, cycle, list, hierarchy or timeline	Structure fits; editability is retained
Charts	Native PowerPoint chart or embedded/linked Excel	Action title, direct labels, units and source are visible

© 2026 Tertiary Infotech Academy Pte Ltd. All rights reserved. · www.tertiarycourses.com.sg

Microsoft 365 Copilot and Gemini Notebook

Microsoft 365 Copilot can create an infographic from a detailed prompt and, where available, an uploaded file. Learners can select brand and colour options, choose Portrait or Wide, generate alternatives and edit or download the result. Copilot in PowerPoint can also create, restructure and refine slides. Availability depends on licence, region, administrator settings and current product rollout.

Gemini Notebook: sources to infographic and PowerPoint

1. Create a notebook and add approved files, websites, pasted text or supported YouTube sources. Select only the sources needed for the brief.
2. In Studio choose Infographic. Use the pencil control to set language, level of detail, orientation, visual style and a custom prompt. Generate, inspect and download the PNG.
3. In Studio choose Slide Deck. Select Presenter Slides for a visual speaking deck or Detailed Deck for a reading document; set language, length and the common prompt.
4. Inspect the generated deck. Use the three-dot menu to Download PowerPoint (.pptx) where that option is available. If the interface exposes only PDF, preserve the PDF and rebuild decision-critical content as native PowerPoint objects; do not assume a PDF conversion is fully editable.
5. Gemini Notebook infographics and decks may contain visual or factual inaccuracies. During slide revisions, sources may not be re-checked, so verify every changed claim again.

Criterion	Microsoft 365 Copilot	Gemini Notebook
Starting point	Prompt or uploaded file in Microsoft 365 Copilot	Selected notebook sources: files, URLs, text or YouTube
Best use	Direct infographic creation and editing within the Microsoft ecosystem	Source-grounded synthesis into an infographic or multi-slide narrative
PowerPoint hand-off	Copy/download or continue editing in Microsoft tools	Download .pptx where available; otherwise preserve PDF and rebuild critical content in PowerPoint
Control	Brand kit, orientation, prompt and visual editor where available	Source selection, format, language, length, orientation/detail/style and custom prompt
Review risk	Availability, editability and generated claims vary by licence and output	Infographics/decks may contain visual or factual errors; revisions may not re-check sources

Human review gate

© 2026 Tertiary Infotech Academy Pte Ltd. All rights reserved. · www.tertiarycourses.com.sg

- Truth — verify every number, label, source and implied relationship.
- Privacy — use only approved data and services; do not upload personal or restricted information.
- Rights — confirm template, image, icon, font and generated-asset permissions.
- Accessibility — check reading order, contrast, direct labels, text size and alt text.
- Brand — apply the approved master, palette, logo, tone and ownership.
- Editability — rebuild decision-critical evidence as native objects and test export.

Review and Resource Decisions

Review an infographic against observed audience outcomes and the six dimensions below. Feedback should describe what is visible, ask about the objective, connect the issue to a principle and propose one testable improvement.

Dimension	Review question
Message clarity	Can the audience state the intended takeaway and action?
Data integrity	Are values, units, context, source and uncertainty accurate?
Visual hierarchy	Is the reading order obvious and purposeful?
Accessibility	Are contrast, labels, text size, alt text and non-colour cues adequate?
Engagement	Does the visual earn attention without chart junk or decoration?
Business and resource fit	Does it support the decision, channel, budget, workflow and governance needs?

LU1 — Introduction to Infographics

Purpose · Types · Audience · Metrics · Planning · Insight extraction

Key concepts

- An infographic combines concise text, data, shapes, colour, charts and icons to reveal information and relationships.
- Chronology, timeline, process, comparison, statistical, mind-map, numerical and hierarchical formats answer different questions.
- An AI-first workflow gathers approved sources, cleans and analyses data, extracts candidate insights, then routes verified evidence into visual generation.
- Insight extraction distinguishes what the chart shows, what it means, why it matters and what action is justified.

Activity 1 — Use AI to Clean Data and Extract Insights

Objective: Use ai to organise and clean approved mock sources, then verify and convert candidate patterns into defensible business insights.

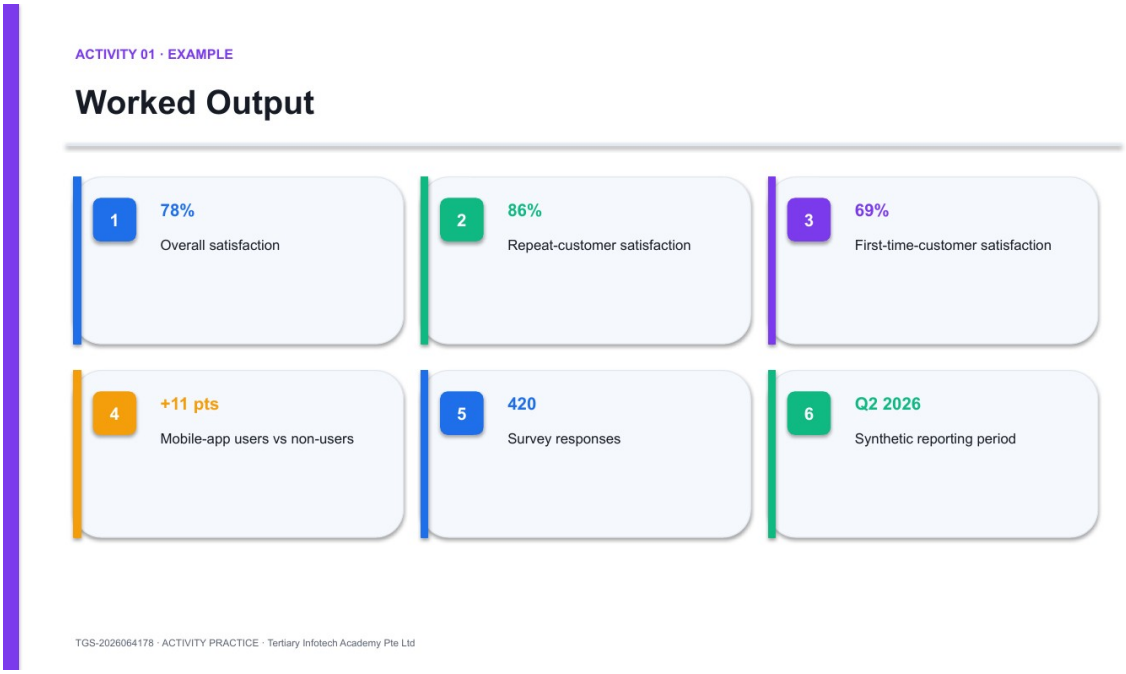
© 2026 Tertiary Infotech Academy Pte Ltd. All rights reserved. · www.tertiarycourses.com.sg

Output: A cleaned analysis workbook, AI prompt/source log, annotated insight slide and one-sentence recommendation (35 minutes).

Scenario

Northstar Retail's marketing lead has a one-page customer survey infographic built from 420 synthetic Q2 responses. Leadership wants to know which customer experience deserves further investigation.

Use an approved AI assistant with the supplied CSV, workbook and source notes to produce a clean analysis and candidate insights. Verify every number in Excel before recommending one next step.



Activity 1 workflow/output preview — use the editable PPTX in the activity folder.

Detailed step-by-step procedure

1. **Register the sources —** Open the A01 XLSX, CSV and Source Notes sheet. In the AI Prompt Lab record each source, its period, data status, owner and permitted use before uploading anything.

2. **Preserve the raw data —** Save a named working copy. Do not modify or replace the raw CSV; perform cleanup on a separate worksheet or in an approved AI session.

3. **Run the data-cleaning prompt —** Copy the Data analyst prompt from PROMPTS.md. Ask the AI to flag duplicates, missing values, inconsistent labels, units and periods and to describe every proposed transformation. Require no invented values.

© 2026 Tertiary Infotech Academy Pte Ltd. All rights reserved. · www.tertiarycourses.com.sg

4. **Verify the cleanup —** Compare the AI response with the raw CSV. Apply only justified changes in the workbook, retain a before/after note and confirm the response count remains 420.

5. **Ask for candidate insights —** Request the largest, smallest, gap, exception and competing explanation. Require each candidate insight to cite the exact rows or source fields used.

6. **Reproduce the calculations —** Use the Analysis sheet and Excel formulas to reproduce every critical difference or percentage. Reject any AI claim that cannot be reproduced.

7. **Write the insight chain —** Complete What it shows; What it may mean; Why it matters; What action is justified. Add one caveat or confidence note.

8. **Build the PowerPoint evidence —** Open the A01 start PPTX, save a working copy, add one editable annotation and show Synthetic Northstar Retail data, Q2 2026, n=420 in the source footer.

9. **Complete the prompt log —** Record the AI tool, full prompt, approved inputs, output reviewed, rejected claims, human changes, reviewer and final owner.

Test it

Change one value in a named workbook copy. Recalculate the insight and confirm the PowerPoint claim, source note and caveat still agree.

Suggested timing

Minutes	Phase
5 min	Review the question and evidence definitions
8 min	Read values and denominators
12 min	Build the insight chain and annotation
7 min	Peer challenge and revision
3 min	Save and submit

Acceptance checklist

- Observation uses visible values and correct units
- Interpretation is labelled as interpretation, not fact
- Source, timeframe and sample size are visible
- Action is proportionate to the evidence
- Editable PPTX opens without repair

Discussion and debrief

- What is the strongest visible pattern?
- Which denominator or segment could change the interpretation?
- What can be said confidently, and what cannot?

© 2026 Tertiary Infotech Academy Pte Ltd. All rights reserved. · www.tertiarycourses.com.sg

- What business action is proportionate to the evidence?

Debrief standard: The final insight clearly separates what the data shows, what it may mean, why it matters and what should happen next.

Activity 2 — Prompt an Evidence-Led Infographic Brief

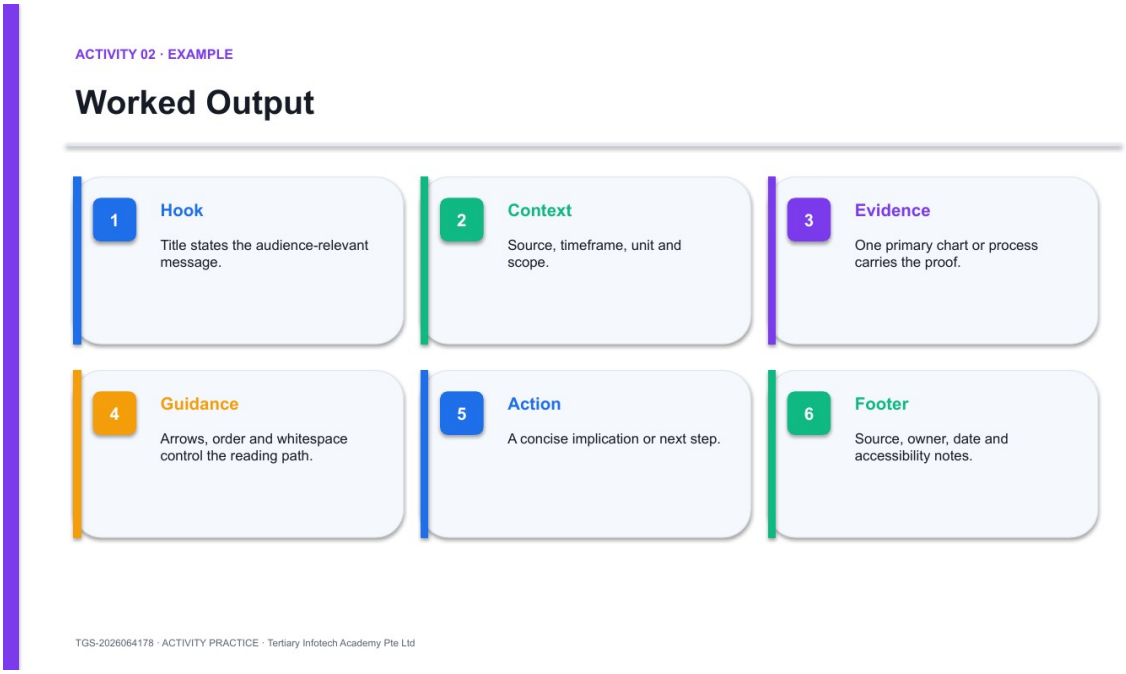
Objective: Turn verified insights into a structured ai brief and prompt that controls audience, message, evidence, form and output.

Output: A structured prompt brief, two AI wireframe directions and one approved PowerPoint wireframe (35 minutes).

Scenario

Northstar plans a sustainable-delivery campaign. The communication manager needs a portrait infographic showing the customer journey from order to low-emission delivery.

Use the supplied prompt workbook to ask AI for two wireframe directions. Select one by evidence fit, then preserve it as the control brief for Microsoft 365 Copilot, Gemini Notebook or PowerPoint.



Activity 2 workflow/output preview — use the editable PPTX in the activity folder.

Detailed step-by-step procedure

1. **Open the brief and prompt pack —** Open the A02 workbook, PROMPTS.md and start PPTX. Save named working copies and select only the verified A01 insight chain as input.

© 2026 Tertiary Infotech Academy Pte Ltd. All rights reserved. · www.tertiarycourses.com.sg

2. **Define the control brief —** Complete Audience, Business question, Desired action, Success evidence, approved claims, forbidden claims, output channel and owner.

3. **Choose the infographic form —** Select timeline, process, comparison, statistical, hierarchy or hybrid. Explain why it reveals the relationship better than two alternatives.

4. **Write the structured prompt —** Use the prompt anatomy in the workbook: role, audience, objective, approved evidence, hierarchy, format, brand, accessibility, constraints and output.

5. **Request two directions —** Ask the approved AI tool for two low-fidelity wireframe directions. Require labelled content zones, source space and no invented data or imagery.

6. **Score the directions —** Compare both options for question fit, hierarchy, evidence space, reading order, accessibility and ease of PowerPoint refinement.

7. **Build the approved wireframe —** In PowerPoint set a portrait canvas, turn on guides and recreate only the selected direction with editable grey placeholders.

8. **Test the reading path —** Zoom to 25 percent. Ask a peer to point to first, second and last. Move or simplify blocks until the intended order is consistent.

9. **Record the decision —** Capture the chosen direction, rejected option, prompt, removed content, intended channel, accessibility risks, reviewer and owner.

Test it

Show the wireframe at 25% zoom for five seconds. Ask a peer to identify the first, second and final reading points without coaching.

Suggested timing

Minutes	Phase
5 min	Complete the brief
5 min	Choose and justify the form
15 min	Build the portrait wireframe
7 min	Test the reading path
3 min	Save and submit

Acceptance checklist

- Audience question and action are explicit
- Infographic form matches the relationship
- Wireframe works without colour
- Reading order is obvious at thumbnail size
- Space exists for source and accessibility notes

Discussion and debrief

© 2026 Tertiary Infotech Academy Pte Ltd. All rights reserved. · www.tertiarycourses.com.sg

- What is the single audience question?
- Which infographic type best reveals the relationship?
- What content should be removed?
- Where should the reader's eye land first, second and last?

Debrief standard: The wireframe communicates objective, hierarchy, evidence, flow, action and source space before visual styling begins.

LU2 — Create Infographics in PowerPoint

Slide size · Grid · Shapes · Icons · Type · Colour · SmartArt · Charts · AI

Key concepts

- Visual hierarchy uses scale, position, contrast, spacing and alignment to guide the reader through one intended path.
- PowerPoint is the editable refinement layer for AI-generated structure, imagery, text, shapes, SmartArt, native charts and Excel-linked data.
- Microsoft 365 Copilot can generate an infographic from a prompt or file; Gemini Notebook can turn selected sources into an infographic or slide deck for PowerPoint refinement.
- Every AI-assisted output needs source verification, privacy review, brand control, accessibility checks and editable hand-off.

Activity 3 — Build from Microsoft Infographic Maker

Objective: Adapt a microsoft infographic template while preserving hierarchy, editability, brand and evidence.

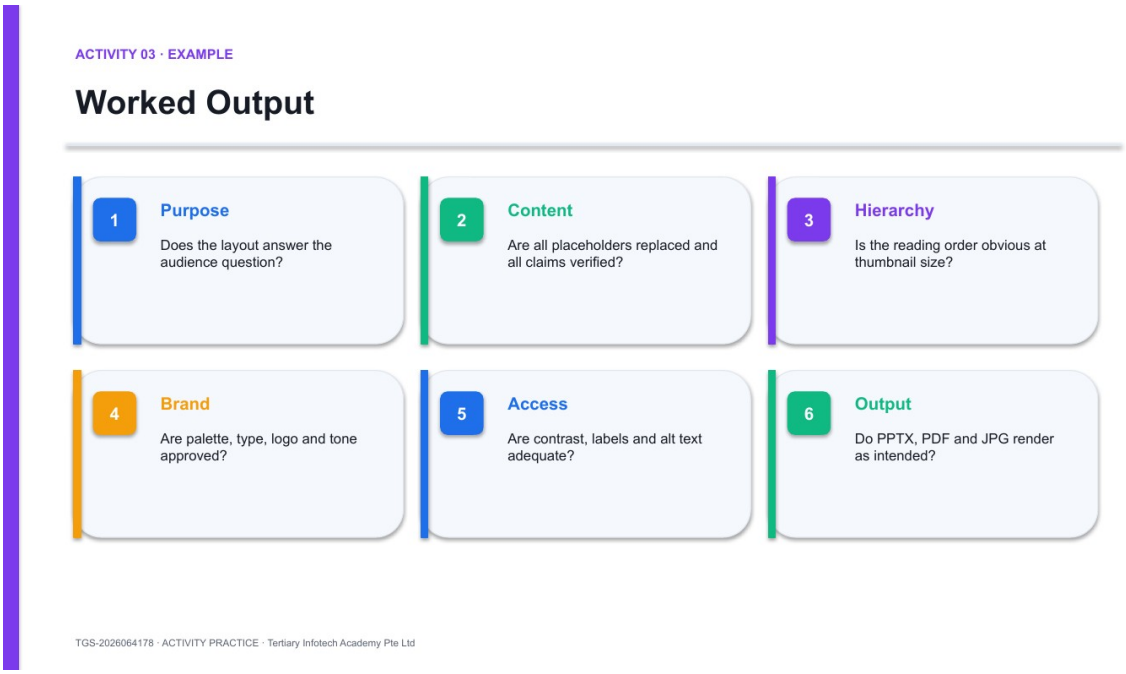
Output: An editable portrait template-based infographic in PPTX, PDF and JPG (40 minutes).

Scenario

Northstar wants a vertical five-stage sustainable-delivery journey for an employee town hall and social post.

Start from a Microsoft infographic-maker template or the supplied editable sample. Adapt it to the approved wireframe; do not let inherited placeholders define the story.

© 2026 Tertiary Infotech Academy Pte Ltd. All rights reserved. · www.tertiarycourses.com.sg



Activity 3 workflow/output preview — use the editable PPTX in the activity folder.

Detailed step-by-step procedure

1. **Open the Microsoft maker —** Open https://powerpoint.cloud.microsoft/create/en/infographic-maker/ in your browser and sign in only with an approved account. If the online maker is unavailable, use the supplied editable start deck.
2. **Choose by relationship —** Select a process, timeline, roadmap or poster template that matches the approved wireframe. Do not choose only because the colours look attractive.
3. **Create a working copy —** Open the template in PowerPoint and immediately use File > Save a Copy. Name it A03-<YourName>-Delivery-Journey.pptx.
4. **Audit every placeholder —** Use Home > Select > Selection Pane. Rename major objects and identify all placeholder text, images, icons, charts, logos and decorative shapes.
5. **Replace the message —** Insert the approved five-stage delivery content from A03-Build-from-Microsoft-Infographic-Maker-Data.xlsx. Delete any template section that has no role in the story.
6. **Apply the approved style —** Use Design > Variants > Colors and Fonts or apply the supplied palette manually. Keep one heading font, one body font and consistent stage colours.

© 2026 Tertiary Infotech Academy Pte Ltd. All rights reserved. · www.tertiarycourses.com.sg

7. **Check alignment —** Select comparable stage objects, then choose Shape Format > Align > Align Middle and Distribute Vertically. Group each stage after alignment.

8. **Add accessibility data —** Right-click meaningful images and choose View Alt Text. Write purpose-focused alt text; mark decorative objects as decorative where supported.

9. **Export and inspect —** Use File > Export > Create PDF/XPS for PDF and File > Save As > PNG for an image. Compare the outputs with the editable master for clipping and font changes.

Test it

Move one stage group and export a fresh PDF. Confirm connectors, labels, hierarchy, source and alt-text decisions remain correct.

Suggested timing

Minutes	Phase
5 min	Template selection and working copy
10 min	Placeholder audit and content replacement
15 min	Brand, hierarchy and alignment
7 min	Accessibility and export
3 min	Save and submit

Acceptance checklist

- Template relationship matches the brief
- No placeholder text remains
- Brand and visual hierarchy are consistent
- Source and alt text are present
- PPTX, PDF and JPG render correctly

Discussion and debrief

- Why does this template fit the question?
- Which inherited element should be removed first?
- What must remain editable?
- Which export format fits each channel?

Debrief standard: The final template-based infographic is message-led, branded, accessible, source-labelled and fully editable.

Activity 4 — Generate Visual Assets and Refine Them in PowerPoint

Objective: Use text-to-image prompting to generate a consistent asset set, then combine it with editable powerpoint labels, shapes, connectors and smartart.

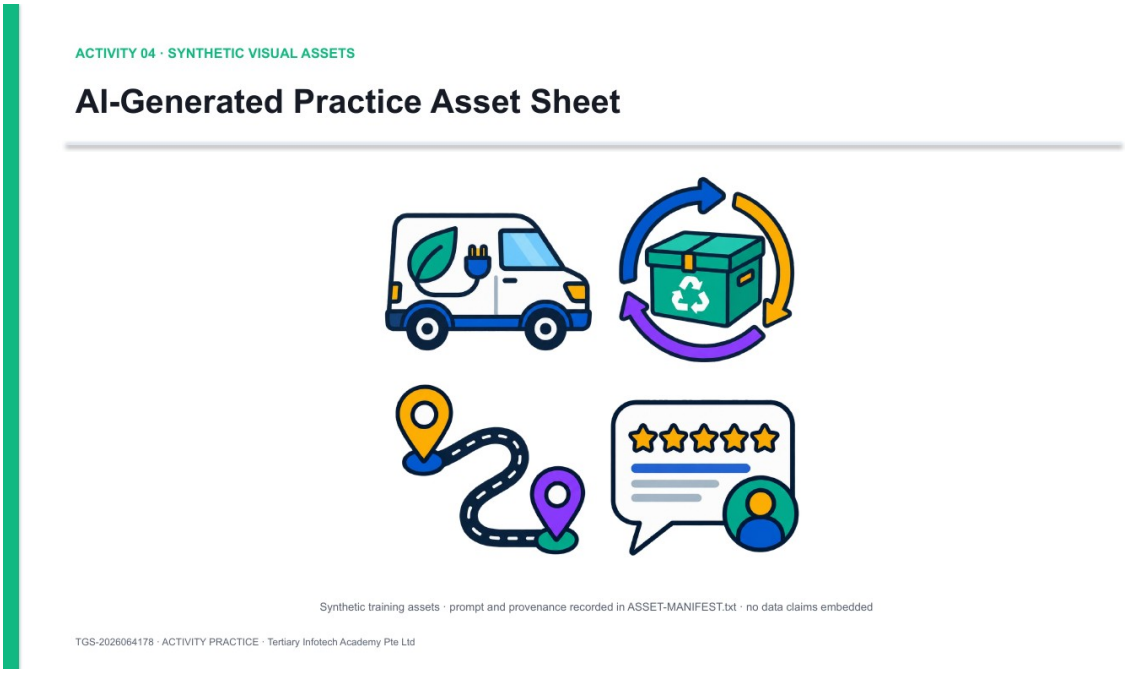
Output: A prompt log, consistent AI-generated asset set, editable five-stage infographic and reusable PowerPoint component slide (35 minutes).

© 2026 Tertiary Infotech Academy Pte Ltd. All rights reserved. · www.tertiarycourses.com.sg

Scenario

Northstar needs an editable five-stage process infographic for store managers. Use the supplied AI-generated sustainability asset sheet or regenerate a new set from the recorded text-to-image prompt.

Use the images only as supporting visuals. Build all stage names, values, connectors, source labels and decision meaning as editable PowerPoint objects.



Activity 4 workflow/output preview — use the editable PPTX in the activity folder.

Detailed step-by-step procedure

1. **Review the supplied assets —** Open the assets folder and ASSET-MANIFEST.txt. Confirm the files are synthetic training visuals, contain no factual claims and match the recorded prompt.

2. **Write or adapt the image prompt —** Copy the Text-to-image prompt from PROMPTS.md. Specify the four sustainability subjects, consistent flat style, palette, transparent background, no text, no logo and no watermark.

3. **Generate an alternative set —** Use an approved text-to-image tool to create one coherent asset sheet or four separate visuals. Do not upload restricted data or copyrighted reference artwork.

4. **Inspect before use —** Reject any result with embedded text, distorted objects, inconsistent visual grammar, ambiguous symbols or rights concerns. Record the accepted and rejected results.

© 2026 Tertiary Infotech Academy Pte Ltd. All rights reserved. · www.tertiarycourses.com.sg

5. **Open the PowerPoint start file —** Save a named working copy, set guides and margins, and insert only the approved assets.
6. **Build editable information —** Create the five stage containers, titles, metrics, connectors, source notes and decision labels as native PowerPoint objects. Do not bake claims into the image.
7. **Align and distribute —** Use Shape Format > Align and Distribute. Group each stage logically while keeping text and data editable.
8. **Add accessibility —** Write purpose-focused alt text for meaningful assets, mark decorative assets where appropriate and preserve direct text labels.
9. **Complete provenance —** Record tool, exact prompt, date, source status, intended use, review result, human changes, reviewer and owner in the workbook and ASSET-MANIFEST.txt.

Test it

Replace one generated asset. Confirm all facts, labels, connectors and sources remain separate, native and editable in PowerPoint.

Suggested timing

Minutes	Phase
5 min	Set grid and build first component
15 min	Duplicate, align and connect
8 min	Insert and format icons
5 min	Compare SmartArt
2 min	Save and submit

Acceptance checklist

- Stages use equal size and spacing
- Connectors remain attached when objects move
- Icons use one family and have direct labels
- The reading path is obvious
- All objects remain editable and grouped logically

Discussion and debrief

- Which visual property signals sequence?
- Where is direct labelling still required?
- Would SmartArt clarify or constrain this design?
- How will the system remain reusable?

Debrief standard: The process uses a consistent component system, real connectors, direct labels, purposeful whitespace and editable PowerPoint objects.

© 2026 Tertiary Infotech Academy Pte Ltd. All rights reserved. · www.tertiarycourses.com.sg

Activity 5 — Use AI Analysis to Build a Native PowerPoint Data Story

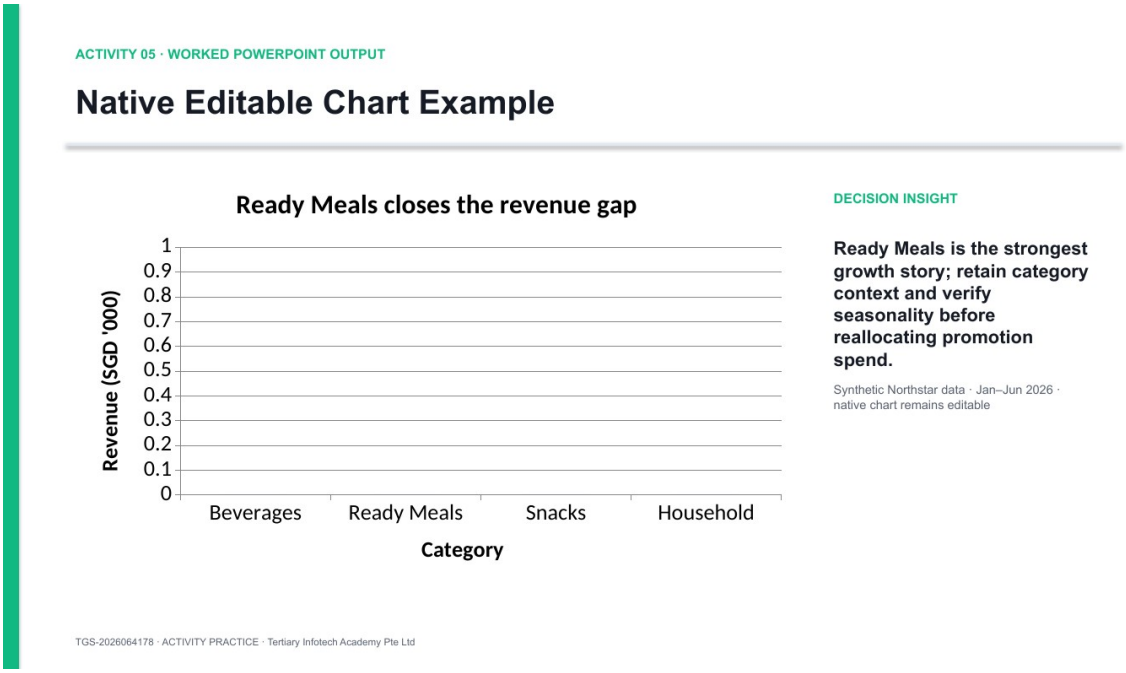
Objective: Use ai to clean and analyse mock excel data, verify calculations, then create a native editable chart and decision story in powerpoint.

Output: An AI analysis log, verified formula workbook, native editable PowerPoint chart and annotated decision slide (45 minutes).

Scenario

Northstar's marketing team must choose one product family for the next promotion. The supplied workbook contains monthly revenue, units, margin and returns for four categories.

Prompt an approved AI assistant to analyse the workbook and propose candidate insights. Reproduce the selected calculations in Excel, then create an editable PowerPoint chart with an explicit decision implication.



Activity 5 workflow/output preview — use the editable PPTX in the activity folder.

Detailed step-by-step procedure

1. **Open and validate the workbook —** Open the A05 XLSX. Review Data, Analysis, Source Notes and AI Prompt Lab. Change one input value and confirm formulas update without errors.
2. **Run the analysis prompt —** Ask an approved AI assistant to flag data-quality issues, compare revenue, growth, margin and returns, calculate candidate metrics and provide citations to the relevant rows. Require no invented values.

© 2026 Tertiary Infotech Academy Pte Ltd. All rights reserved. · www.tertiarycourses.com.sg

3. **Verify every calculation —** Reproduce the selected growth, totals, averages and comparisons with Excel formulas. Record any rejected AI claim and why.

4. **Define the decision metric —** State which product family to promote and the minimum evidence required. Separate observed patterns from hypotheses and caveats.

5. **Choose the chart —** Use a clustered bar for category comparison or a line for change over time. Explain why the chart reveals the relationship better than a pie or 3-D chart.

6. **Create the Excel chart —** Select the prepared Analysis range, choose Insert > Recommended Charts, select the intended chart, and add a precise chart title.

7. **Paste as a native chart —** Copy the chart. In PowerPoint choose Home > Paste > Use Destination Theme & Embed Workbook or Keep Source Formatting & Link Data. Do not paste as a picture.

8. **Remove chart junk —** Use Chart Design > Add Chart Element. Keep only necessary axes, direct labels and a short title. Remove background fills, 3-D effects, unnecessary gridlines and legends.

9. **Write the action title —** Replace the topic title with a sentence that states the supported conclusion, for example Ready Meals combine the fastest Q2 growth with above-floor margin.

10. **Highlight the decision —** Use one accent colour for the selected category. Keep all comparison categories neutral and add one concise annotation.

11. **Add source and limitation —** Show Synthetic Northstar data, Jan-Jun 2026 and one relevant caveat such as promotional seasonality or return-rate uncertainty.

12. **Test refresh and log —** Change one workbook input, refresh the chart and confirm the visual updates. Complete the AI prompt/source log with tool, prompt, output, verification and owner.

Test it

Change one Excel input, refresh the native chart and verify the action title, direct labels, unit and limitation against the updated data.

Suggested timing

Minutes	Phase
8 min	Validate data and define metrics
10 min	Build the Excel chart
15 min	Paste and format the native chart
8 min	Write insight, source and limitation
4 min	Refresh test and submit

Acceptance checklist

© 2026 Tertiary Infotech Academy Pte Ltd. All rights reserved. · www.tertiarycourses.com.sg

- Workbook has zero formula errors
- Chart type matches the relationship
- Action title states a defensible insight
- One accent highlights the decision
- Source, timeframe, unit and limitation are visible

Discussion and debrief

- Which metric answers the actual business question?
- What does the chart show that a table does not?
- Which context or caveat belongs on the slide?
- Which product should be tested and why?

Debrief standard: The chart is native/editable, uses an appropriate form, retains source context and makes a proportionate product-promotion recommendation.

Activity 6 — Compare Microsoft 365 Copilot and Gemini Notebook

Objective: Generate the same source-grounded infographic brief in microsoft 365 copilot and gemini notebook, bring the gemini notebook deck into powerpoint through pptx download or a documented pdf rebuild route, and compare both routes before refinement.

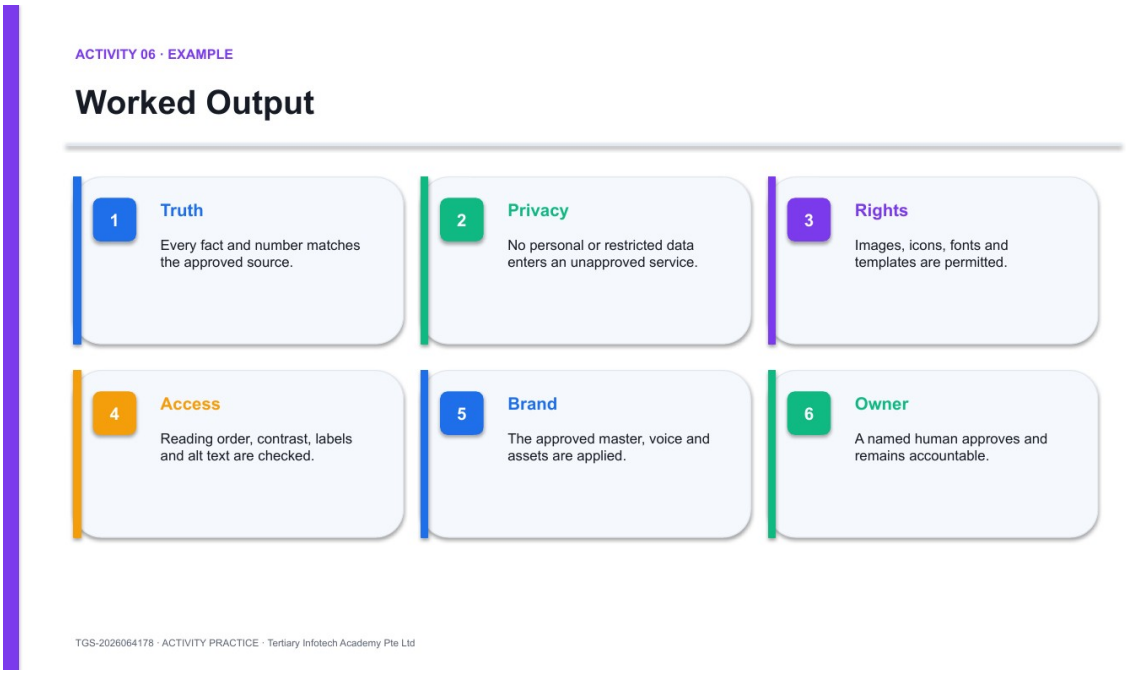
Output: A Microsoft output, Gemini Notebook infographic and downloaded PPTX or PDF, comparison scorecard, reviewed final PowerPoint and prompt/source log (45 minutes).

Scenario

Northstar wants an executive infographic and short slide narrative summarising the sustainable-delivery pilot from a spreadsheet, policy note and stakeholder feedback summary.

Run the same verified brief through Microsoft 365 Copilot and Gemini Notebook. Generate a Gemini Notebook Infographic plus Presenter Slides, preserve the deck as PPTX where available or PDF with a documented rebuild route, compare both routes and refine the selected master in PowerPoint.

© 2026 Tertiary Infotech Academy Pte Ltd. All rights reserved. · www.tertiarycourses.com.sg



Activity 6 workflow/output preview — use the editable PPTX in the activity folder.

Detailed step-by-step procedure

1. **Open the common brief —** Open the A06 workbook, PROMPTS.md and source-pack folder. Complete the audience, objective, approved sources, verified claims, forbidden claims, infographic form, brand and accessibility rules.
2. **Run the Microsoft route —** In the Microsoft 365 Copilot app choose Create > Design an infographic, or use Copilot in PowerPoint. Add the approved file where available, submit the common prompt and save the generated option.
3. **Run the Gemini Notebook route —** Create a notebook, add only the approved source-pack files and select them. In Studio customise and generate an Infographic using the same audience, focus, style, orientation and evidence constraints.
4. **Generate Gemini Notebook slides —** In Studio select Slide Deck, choose Presenter Slides, set the language and length and reuse the common prompt. Generate and inspect the deck. Use the three-dot menu to Download PowerPoint (.pptx) when that option is available. If the interface exposes only PDF, download the PDF, keep it as evidence and rebuild decision-critical content as native PowerPoint objects rather than assuming the PDF is fully editable.
5. **Preserve originals —** Keep the Microsoft output, Gemini Notebook infographic PNG and downloaded Gemini Notebook PPTX or PDF unchanged as evidence. Work only on named copies.

© 2026 Tertiary Infotech Academy Pte Ltd. All rights reserved. · www.tertiarycourses.com.sg

6. **Compare the routes —** Use the Comparison sheet to score source grounding, factual accuracy, citations, hierarchy, visual quality, editability, brand, accessibility, speed and privacy fit.

7. **Verify every claim —** Compare every number, label and relationship with the workbook and source pack. Note that AI-generated infographics and decks may contain factual or visual inaccuracies.

8. **Refine in PowerPoint —** Choose the stronger direction. Rebuild decision-critical numbers, text, labels and charts as native objects; apply brand, source footer, alt text and correct reading order.

9. **Record ownership —** Complete the prompt/source log with both tools, selected sources, prompts, outputs, rejected claims, human changes, reviewer and final owner.

Test it

Select one claim produced by both AI routes. Trace it to the same approved source and document any difference in wording, hierarchy or editability.

Suggested timing

Minutes	Phase
7 min	Complete the structured prompt
8 min	Generate two options
10 min	Score and verify
15 min	Rebuild and check accessibility
5 min	Complete log and submit

Acceptance checklist

- Both tools used the same approved brief and source scope
- Gemini Notebook deck was preserved as PPTX where available or PDF with a documented rebuild route
- Every claim matches the source pack
- Critical evidence is editable in the final PowerPoint
- Both prompts, human changes, reviewer and owner are recorded

Discussion and debrief

- What must the prompt include to prevent a generic output?
- Which data cannot be sent to an unapproved service?
- What must be rebuilt as native PowerPoint objects?
- Who signs off the final infographic?

Debrief standard: The final slide records the prompt, sources and human changes; all claims are verified and the decision evidence remains editable.

© 2026 Tertiary Infotech Academy Pte Ltd. All rights reserved. · www.tertiarycourses.com.sg

LU3 — Review Infographics

Story · Context · Simplicity · Coaching · Budget · Effectiveness · Improvement

Key concepts

- Effective review separates message clarity, data integrity, visual hierarchy, accessibility, audience fit and business usefulness.
- Coaching uses specific observations, questions and criteria rather than taste-based comments such as 'make it pop'.
- Tool investment is evaluated through capability, licence, learning, production, governance, collaboration and maintenance costs.
- Improvement is evidence-led: gather feedback, prioritise changes, revise the controlled master and retest the intended outcome.

Activity 7 — Peer Coaching and Story Review

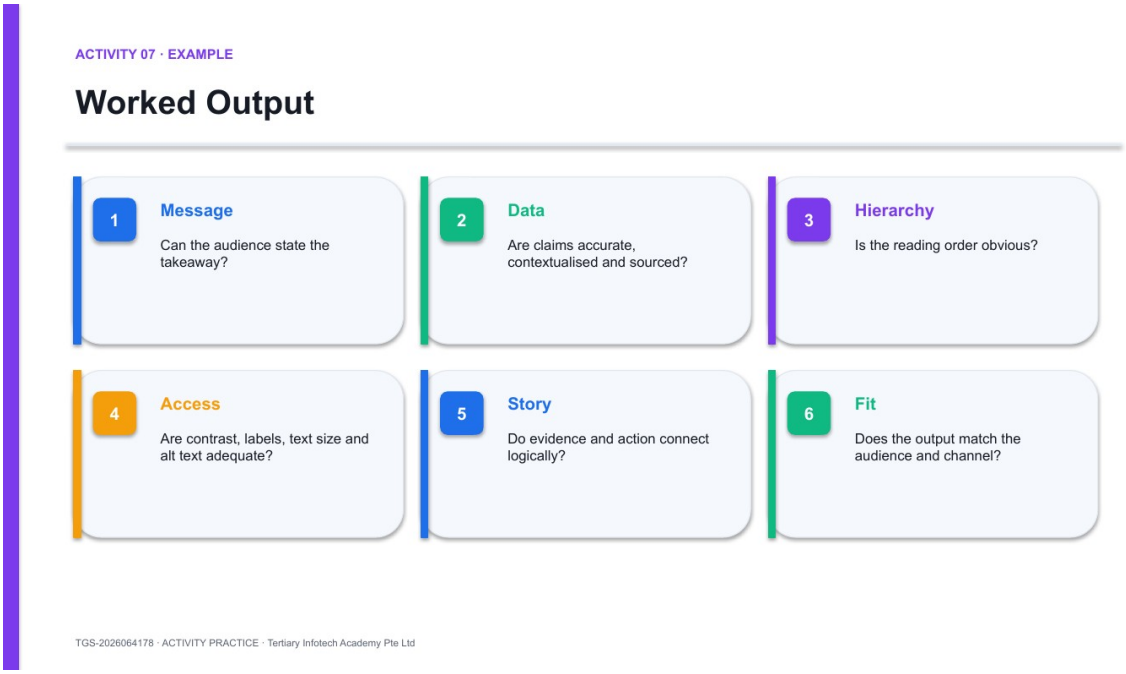
Objective: Give specific criteria-based feedback and revise an infographic to improve clarity, story and accessibility.

Output: A completed peer-review log, revised infographic and short reflection (30 minutes).

Scenario

Each learner presents one infographic from Activities 3–6 to two peers. The group must distinguish aesthetic preference from evidence-based review.

Capture two observations, two strengths, two improvements and one agreed revision. Then revise the master and record the change.



© 2026 Tertiary Infotech Academy Pte Ltd. All rights reserved. · www.tertiarycourses.com.sg

Activity 7 workflow/output preview — use the editable PPTX in the activity folder.

Roles

- Designer — Goal: explain the intended audience and message. Listen without defending every choice; ask for evidence.
- Reviewer — Goal: give criteria-based feedback. Use observations, questions and one prioritised suggestion.
- Audience representative — Goal: test comprehension. State what you understood, what confused you and what action you would take.

Detailed step-by-step procedure

1. **Choose the review slide —** Open one infographic from Activities 3-6 and save a copy for revision.
2. **Assign roles —** Work in groups of three: Designer, Reviewer and Audience representative. Rotate roles if time permits.
3. **Run the five-second test —** Show the slide for five seconds, then hide it. The audience representative states the message, evidence and intended action they remember.
4. **Record observations —** In A07-Peer-Coaching-and-Story-Review-Data.xlsx capture visible facts only: reading order, text size, labels, contrast, source, chart comprehension and action clarity.
5. **Ask clarifying questions —** The reviewer asks which audience, message, evidence and constraint drove the design. The designer answers without dismissing the observation.
6. **Apply the six-lens rubric —** Score Message, Data, Hierarchy, Accessibility, Story and Channel fit. Add one evidence note for every score.
7. **Prioritise one change —** Choose the issue with the largest audience or decision impact. State the change, owner and acceptance check.
8. **Revise the master —** Update the editable PPTX. Do not flatten the file or create an uncontrolled duplicate.
9. **Retest and reflect —** Repeat the five-second test and write 150-250 words on what improved, what remains and why.

Test it

Repeat the five-second audience test before and after revision. Record the observed change in recall, reading order or action clarity.

Suggested timing

© 2026 Tertiary Infotech Academy Pte Ltd. All rights reserved. · www.tertiarycourses.com.sg

Minutes	Phase
5 min	Assign roles and run five-second test
10 min	Observe, question and score
8 min	Prioritise and revise
5 min	Retest
2 min	Save reflection

Acceptance checklist

- Feedback contains observations, not taste-only comments
- Every score has evidence
- One high-impact change is prioritised
- Revised PPTX remains editable
- Reflection explains how the change affects audience understanding

Discussion and debrief

- What did the audience understand first?
- Which issue has the greatest decision impact?
- Which feedback is evidence-based rather than taste?
- How will the change be tested?

Debrief standard: The review log contains observed evidence, a principle-based diagnosis, one prioritised change, a named owner and an acceptance check.

Activity 8 — Budget and Effectiveness Decision

Objective: Compare infographic-tool investment options and recommend a resource plan using effectiveness and governance criteria.

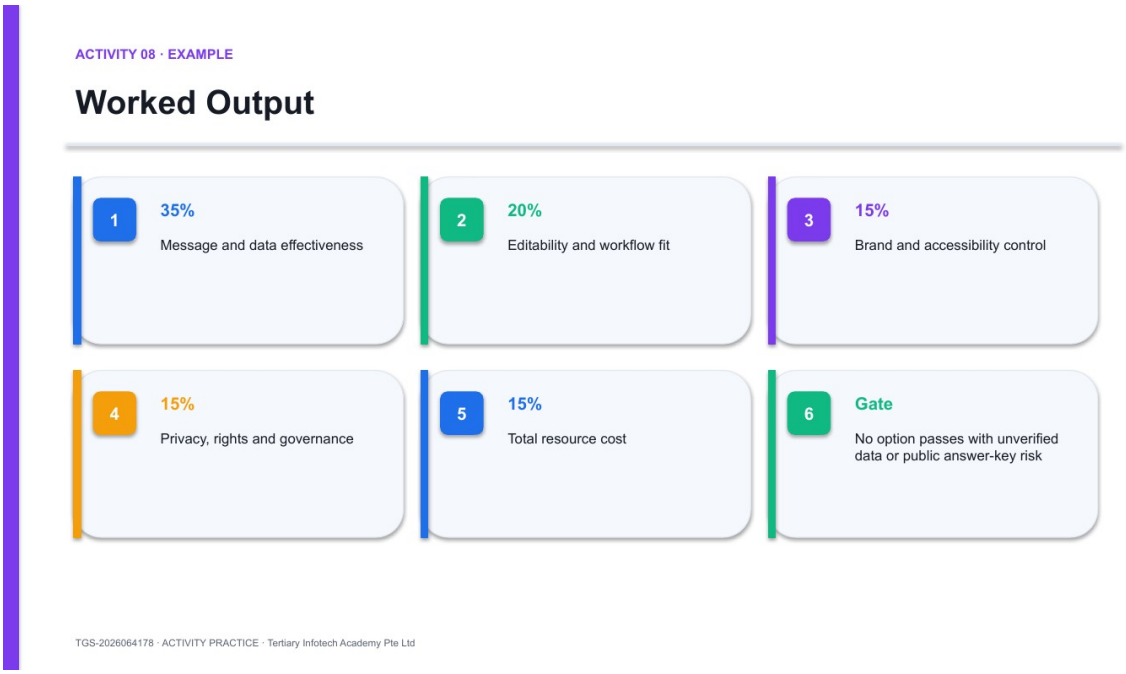
Output: A weighted tool-investment scorecard and one-page resource recommendation (30 minutes).

Scenario

Northstar can fund one standard workflow for 12 staff. Options include PowerPoint templates only, PowerPoint with Copilot, Gemini Notebook with PowerPoint, or a mixed AI workflow with PowerPoint as the controlled final-production layer.

Use the supplied budget workbook and effectiveness evidence to recommend a 12-month tool, training and governance plan aligned with business outcomes.

© 2026 Tertiary Infotech Academy Pte Ltd. All rights reserved. · www.tertiarycourses.com.sg



Activity 8 workflow/output preview — use the editable PPTX in the activity folder.

Detailed step-by-step procedure

1. **Open the options workbook —** Open A08-Budget-and-Effectiveness-Decision-Data.xlsx and review Assumptions, Options, Scorecard and Recommendation.
2. **Confirm the business need —** Record the number of users, infographic volume, required channels, approved data classification and business outcome.
3. **Review total resource cost —** Check licence, training, production, review, governance, maintenance and estimated rework costs. Do not compare licence fees alone.
4. **Set decision weights —** Use the supplied weights or adjust them with a written rationale. Keep effectiveness and hard governance gates visible.
5. **Score each workflow —** Score PowerPoint templates, PowerPoint + Copilot, Gemini Notebook + PowerPoint and Mixed AI tools + PowerPoint using the same scale and evidence.
6. **Test hard gates —** Mark any option that fails data handling, rights, editability, accessibility or continuity as Not eligible regardless of weighted score.
7. **Review formulas —** Confirm weighted totals and 12-month costs recalculate. Resolve any #REF!, #DIV/0!, #VALUE!, #N/A or #NAME? error before proceeding.

© 2026 Tertiary Infotech Academy Pte Ltd. All rights reserved. · www.tertiarycourses.com.sg

8. **Write the recommendation —** In the Recommendation sheet name the preferred workflow, 12-month budget, training plan, governance controls, owner, success metrics and review date.

9. **Create the decision slide —** Open A08-Budget-and-Effectiveness-Decision-Start.pptx and add a native comparison chart, key trade-offs and one explicit recommendation.

10. **Challenge the decision —** Ask which evidence could reverse the choice. Record the trigger and the date for the formal review.

Test it

Change one decision weight or hard-gate result. Confirm eligibility, weighted score, 12-month cost and recommendation respond as intended.

Suggested timing

Minutes	Phase
5 min	Confirm need and assumptions
10 min	Review cost and score options
7 min	Apply hard gates and validate formulas
6 min	Write recommendation and decision slide
2 min	Challenge and submit

Acceptance checklist

- All options use the same criteria and scale
- Total cost includes hidden resource costs
- Governance gates override weighted score
- Workbook has zero formula errors
- Recommendation names budget, controls, metrics, owner and review date

Discussion and debrief

- Which business outcomes define effectiveness?
- What hidden resource costs could reverse the decision?
- Which risks are hard gates rather than weighted preferences?
- When should the investment be reviewed?

Debrief standard: The recommendation names the preferred workflow, budget, ownership, governance controls, success metrics and review date.

Quick PowerPoint Reference

Task	Ribbon path / method
Change orientation	Design > Slide Size > Custom Slide Size
Show layout aids	View > Ruler, Gridlines, Guides
Align and distribute	Select objects > Shape Format > Align

© 2026 Tertiary Infotech Academy Pte Ltd. All rights reserved. · www.tertiarycourses.com.sg

Manage object order	Home > Select > Selection Pane
Insert icons	Insert > Icons
Insert SmartArt	Insert > SmartArt
Create chart	Insert > Chart, or copy from Excel and choose an embed/link paste option
Add alt text	Right-click object > View Alt Text
Check accessibility	Review > Check Accessibility
Export PDF / image	File > Export or File > Save As

Assessment

- Written Assessment (WA-SAQ) — nine open-ended questions, 30 minutes, open book.
- Practical Performance (PP) — three progressive tasks, 30 minutes, open book.
- Only the course slides, this Learner Guide and approved materials may be used.
- Grading: Competent / Not Yet Competent.
- A minimum of 75% attendance is required. Learners must be assessed as Competent and complete the required digital attendance and TRAQOM steps.

Assessment Flow

6. TRAQOM — access the required survey from the LMS.
7. Complete Assessment Digital Attendance.
8. Sit the Written Assessment (WA-SAQ), then Practical Performance (PP).
9. Submit the completed assessment answers on the LMS.
10. Sign the Assessment Summary Record.

LMS: https://lms-tms.tertiaryinfotech.com/

Sources and Further Reading

- Microsoft Support: Create an infographic with Microsoft 365 Copilot: https://support.microsoft.com/en-US/Microsoft-365-Copilot/create-an-infographic-with-the-microsoft-365-copilot-app
- Microsoft Support: Edit with Copilot in PowerPoint: https://support.microsoft.com/en-US/PowerPoint/edit-with-copilot-in-powerpoint
- Google Gemini Notebook Help: Generate an Infographic: https://support.google.com/notebooklm/answer/16758265?hl=en
- Google Gemini Notebook Help: Generate a Slide Deck: https://support.google.com/notebooklm/answer/16757456?hl=en
- Analytics Vidhya: Infographics and PPTs with NotebookLM (now Gemini Notebook): https://www.analyticsvidhya.com/blog/2025/12/infographics-and-ppts-with-notebooklm/

© 2026 Tertiary Infotech Academy Pte Ltd. All rights reserved. · www.tertiarycourses.com.sg

- Alex McFarland: Create infographics with NotebookLM (now Gemini Notebook): https://alexmcfarland.substack.com/p/how-to-create-stunning-infographics
- Paul J Lipsky: NotebookLM infographic style prompts (now Gemini Notebook): https://www.youtube.com/watch?v=2rKAAf_LRVo
- Microsoft Support: Add or remove chart titles: https://support.microsoft.com/office/add-or-remove-titles-in-a-chart-4cf3c009-1482-4908-922a-997c32ea8250
- Microsoft Support: Add alternative text: https://support.microsoft.com/office/add-alternative-text-to-a-shape-picture-chart-smartart-graphic-or-other-object-44989b2a-903c-4d9a-b742-6a75b451c669
- Microsoft Support: Make presentations accessible: https://support.microsoft.com/office/make-your-powerpoint-presentations-accessible-to-people-with-disabilities-6f7772b2-2f33-4bd2-8ca7-dae3b2b3ef25
- Microsoft Support: Change slide size: https://support.microsoft.com/office/change-the-size-of-your-slides-040a811c-be43-40b9-8d04-0de5ed79987e
- Microsoft Support: Align or arrange objects: https://support.microsoft.com/office/align-or-arrange-objects-bfd91078-2078-4b35-8672-f6270690b3b8
- Microsoft Support: Insert icons: https://support.microsoft.com/office/insert-icons-in-microsoft-365-e2459f17-3996-4795-996e-b9a13486fa79
- Microsoft Support: Use charts and graphs: https://support.microsoft.com/office/use-charts-and-graphs-in-your-presentation-c74616f1-a5b2-4a37-8695-fbcc043bf526
- Microsoft Support: SmartArt graphics: https://support.microsoft.com/office/learn-more-about-smartart-graphics-6ea4fdb0-aa40-4fa9-9348-662d8af6ca2c
- W3C: Understanding Success Criterion 1.4.3 Contrast: https://www.w3.org/WAI/WCAG22/Understanding/contrast-minimum.html
- Datawrapper Academy: What to consider when creating charts: https://academy.datawrapper.de/article/127-what-to-consider-when-creating-charts

Support

- Email: enquiry@tertiaryinfotech.com
- Tel: +65 6100 0613
- Website: www.tertiarycourses.com.sg
- LMS: https://lms-tms.tertiaryinfotech.com/

© 2026 Tertiary Infotech Academy Pte Ltd. All rights reserved. · www.tertiarycourses.com.sg